

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Amazon Web Services, OH -- station KOSU, America/New_York
Committed pair OSM 821081357 -> 1154503132
Amazon Web Services / Amazon Web Services
Facade gap 143.0 m between the two halls
Weather record 43,575 real hourly records from KOSU
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3496 C at 0 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2024-02-28 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 16 of 24 hours with 1 mode change. That is 10 more than the reactive incumbent, at 0 unsafe hours against its 2. 6 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 16 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 6 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s)
6 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	16.299	18.0	16.110	1.030
01	mechanical	yes	switch budget	16.794	18.0	16.670	1.155
02	mechanical	yes	switch budget	16.799	18.0	16.670	1.086
03	mechanical	yes	switch budget	17.944	18.0	18.890	1.048
04	mechanical	no	dry-bulb	18.489	18.0	19.440	0.962
05	mechanical	yes	-	16.247	18.0	15.000	0.873
06	mechanical	yes	-	17.367	18.0	15.000	0.775
07	mechanical	no	dry-bulb	19.088	18.0	15.560	1.220
08	FREE	yes	-	16.773	18.0	13.330	1.673
09	FREE	yes	-	15.408	18.0	7.784	2.260
10	FREE	yes	-	15.460	18.0	6.110	2.489
11	FREE	yes	-	12.969	18.0	4.440	2.099

12	FREE	yes	-	8.621	18.0	2.780	1.615
13	FREE	yes	-	7.073	18.0	2.780	1.360
14	FREE	yes	-	5.380	18.0	2.220	1.127
15	FREE	yes	-	3.704	18.0	1.670	0.925
16	FREE	yes	-	3.280	18.0	1.110	1.174
17	FREE	yes	-	1.827	18.0	0.000	1.071
18	FREE	yes	-	0.962	18.0	-1.103	1.289
19	FREE	yes	-	-0.007	18.0	-1.633	1.367
20	FREE	yes	-	-0.672	18.0	-2.207	1.615
21	FREE	yes	-	-1.646	18.0	-3.202	1.587
22	FREE	yes	-	-2.273	18.0	-3.322	1.383
23	FREE	yes	-	-2.892	18.0	-3.890	1.134

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.299 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.794 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.799 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.944 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.489 C against a 18.0 C limit, so it fails by 0.489 C. It would take a limit 0.489 C higher, or a bound 0.489 C tighter, to change this hour.

05:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.247 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.367 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.088 C against a 18.0 C limit, so it fails by 1.088 C. It would take a limit 1.088 C higher, or a bound 1.088 C tighter, to change this hour.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.773 C, which is 1.227 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.673 C, and the margin is measured, not chosen: 1.564 C of group-conditional forecast error for this hour of day, plus 0.1085 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.408 C, which is 2.592 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.260 C, and the margin is measured, not chosen: 2.131 C of group-conditional forecast error for this hour of day, plus 0.1283 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.460 C, which is 2.540 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.489 C, and the margin is measured, not chosen: 2.303 C of group-conditional forecast error for this hour of day, plus 0.1852 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.969 C, which is 5.031 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.099 C, and the margin is measured, not chosen: 1.905 C of group-conditional forecast error for this hour of day, plus 0.1935 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

12:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.621 C, which is 9.379 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.616 C, and the margin is measured, not chosen: 1.420 C of group-conditional forecast error for this hour of day, plus 0.1958 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

13:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.073 C, which is 10.927 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.360 C, and the margin is measured, not chosen: 1.232 C of group-conditional forecast error for this hour of day, plus 0.1282 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.380 C, which is 12.620 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.127 C, and the margin is measured, not chosen: 1.022 C of group-conditional forecast error for this hour of day, plus 0.1050 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 3.704 C, which is 14.296 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.925 C, and the margin is measured, not chosen: 0.837 C of group-conditional forecast error for

this hour of day, plus 0.0886 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 3.280 C, which is 14.720 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.174 C, and the margin is measured, not chosen: 0.997 C of group-conditional forecast error for this hour of day, plus 0.1770 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 1.827 C, which is 16.173 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.071 C, and the margin is measured, not chosen: 0.909 C of group-conditional forecast error for this hour of day, plus 0.1624 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.962 C, which is 17.038 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.289 C, and the margin is measured, not chosen: 1.068 C of group-conditional forecast error for this hour of day, plus 0.2209 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.007 C, which is 18.007 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.367 C, and the margin is measured, not chosen: 1.159 C of group-conditional forecast error for this hour of day, plus 0.2084 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.672 C, which is 18.672 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.615 C, and the margin is measured, not chosen: 1.381 C of group-conditional forecast error for this hour of day, plus 0.2345 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -1.646 C, which is 19.646 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.587 C, and the margin is measured, not chosen: 1.361 C of group-conditional forecast error for this hour of day, plus 0.2262 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -2.273 C, which is 20.273 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.383 C, and the margin is measured, not chosen: 1.185 C of group-conditional forecast error for this hour of day, plus 0.1985 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -2.892 C, which is 20.892 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.134 C, and the margin is measured, not chosen: 0.971 C of group-conditional forecast error for this hour of day, plus 0.1629 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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